

- 6 Guard cells are located in the epidermis of the leaves of most plants. When environmental conditions change, this causes changes in guard cells that control the opening and closing of stomata.

(a) Describe the structure of guard cells.

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..... [4]

(b) Stomata have daily rhythms of opening and closing.

Fig. 6.1 shows the percentage of stomata open at different times of the day, over a period of three days, in the thale cress plant, *Arabidopsis thaliana*.

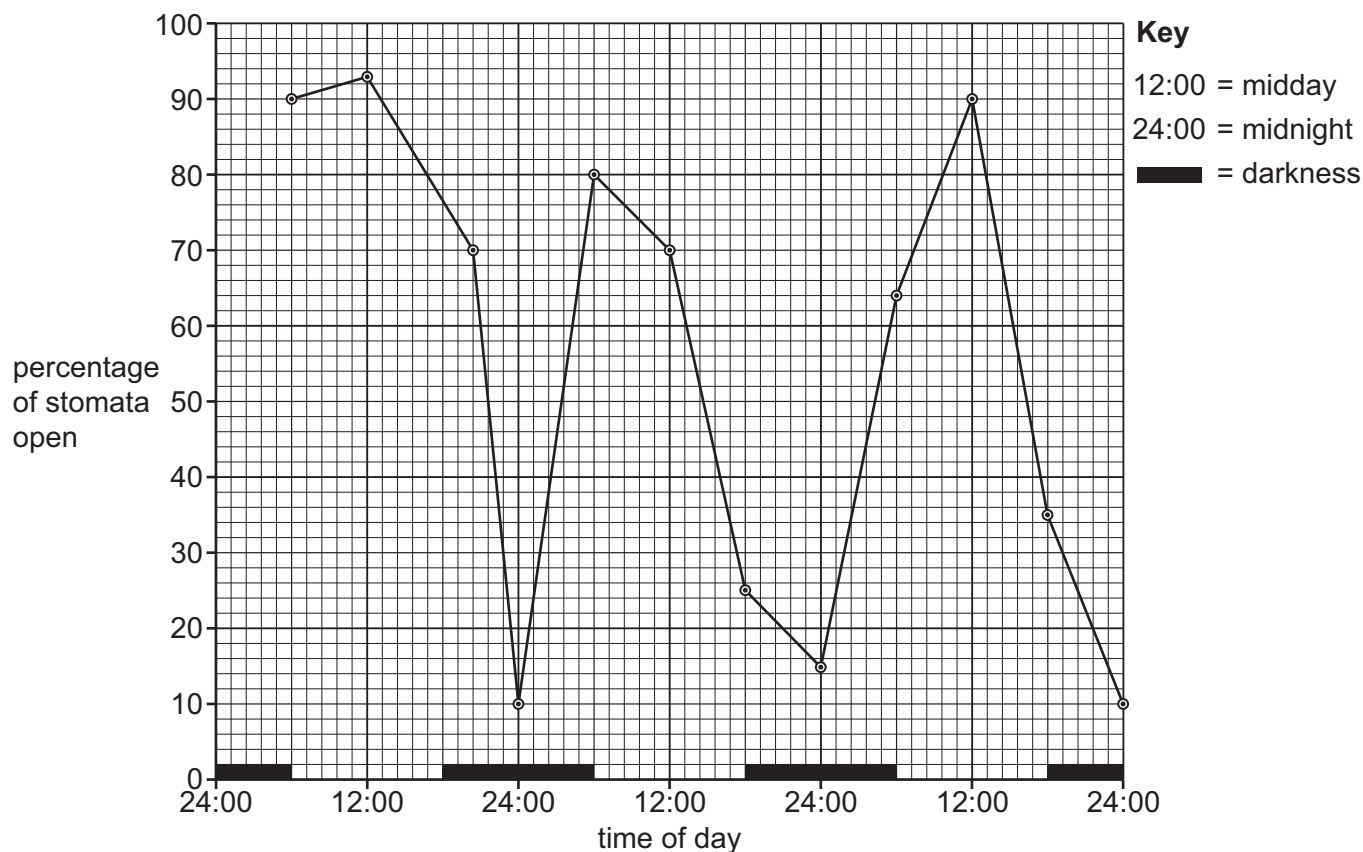


Fig. 6.1



(i) With reference to Fig. 6.1, describe the rhythm of stomatal opening and closing.

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..... [2]

(ii) Suggest other environmental factors, apart from the time of day, that can contribute to stomatal closure.

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..... [2]

(c) Table 6.1 shows some of the events occurring during the closure of a stoma.

The events are **not** listed in the correct order.

Table 6.1

event	description of event
A	water leaves the guard cells by osmosis
B	active transport of hydrogen ions out of the guard cells stops
C	stoma closes
D	plant is subjected to a change in environmental conditions
E	plant releases abscisic acid
F	calcium ions move into the cytoplasm of the guard cells
G	abscisic acid binds to receptors on the cell surface membrane of guard cells
H	guard cells become flaccid
I	water potential of the guard cells increases
J	potassium ions leave the guard cells

Complete Table 6.2 to show the correct order of the events shown in Table 6.1.

Three of the events have been completed for you.

Table 6.2

correct order	letter of event
1	D
2	
3	
4	
5	F
6	
7	
8	
9	
10	C

[4]

[Total: 12]